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ORIGINAL ARTICLE

# Effect of temperature on crystallite size of hydroxyapatite powders obtained by wet precipitation process

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Crystallite size

**Abstract** In this investigation, HAp powders were synthesized using the wet chemical precipitation technique. The temperature of the heat treatment (80 °C, 120 °C, and 160 °C) and the addition of glutamic acid were the considered process parameters. After the reaction between the precursors calcium nitrate [$Ca(NO_3)_2$] and ammonium phosphate [$(NH_4)H_2PO_4$], decantation of the residue, drying, and finally, heat treatment of the residue were done sequentially. X-ray diffraction (XRD) analysis, scanning electron microscope (SEM) observations, and X-ray fluorescence (XRF) analysis were carried out to characterize the synthesized HAp powders. It was found that at a high heat treatment temperature plus the addition of glutamic acid are suitable process parameters to acquire uniform HAp powders with plate morphology and fibers with an average particle size of ~100–200 μm. The Ca/P ratio obtained was like the hydroxyapatite present in the bones in the order of 1.72. This situation can be indicated as an essential advantage in the biocompatibility of the synthesized material. The use of glutamic acid suggests crystal growth in a preferential direction as reported in our previous work. The manufacture of hydroxyapatite, especially in powder, is of great interest in developing additive manufacturing systems for the biomedical market.

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## 1. Introduction

Hydroxyapatite is the most widely used calcium phosphate in biomedical applications due to its high biocompatibility and similarity to human bone structure. Hydroxyapatite is the fundamental constituent of the inorganic part of hard tissues such

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as bones and teeth, can be used as a coating or filling powder for bone fractures, etc. [1,2]. Bones contain 65 wt% of needle-shaped hydroxyapatite with a width of 5–20 nm and length of 60 nm, which is responsible for strength and stiffness, while the organic part (collagen type 1) provides elasticity [3,4].

Several methods have been developed in the past to produce hydroxyapatite powder, including wet synthesis, such as direct precipitation from aqueous solutions, electrochemical deposition [5], solid-state reactions, sol–gel processes [6], and hydrothermal synthesis [7-9]. All hydroxyapatite synthesis processes are intended to obtain materials with specific characteristics such as surface area, grain size, and morphology. Wet chemical precipitation is the most used technique in terms of simplicity and cost and the low working temperature and high percentages of pure product that can be obtained [10-12]. As indicated in the literature, the wet chemical precipitation technique depends on the mechanism of growth-aglomeration-nu cleation-aggregation-agglomeration. The procedure begins with preparing alkaline (Ca source) and acid (P source) solutions and the reaction between them. It proceeds with the stages of precipitation, filtration, drying, and finally heat treatment, respectively [13-15]. Process parameters such as reaction temperature, reagent concentration and drying conditions play an important role in nanostructure morphology and stoichiometry. These parameters must be controlled to achieve the nanostructure's reproducibility.

The synthesis of biocompatible powders with the appropriate particle size and shape is generating significant interest in preparing devices for biomedical demands [12]. For this purpose, the current study focuses on obtaining hydroxyapatite powders following the wet chemical precipitation method. The present work reports the synthesis of hydroxyapatite powders. This study also presents the effects of synthesis parameters such as reaction temperature and drying. Effects of synthesis parameters on the chemical, morphological and structural properties were evaluated according to the results of the characterization analyzes. The goal of this work is to highlight the importance of the use of glutamic acid in the wet precipitation process, which has not been reported in similar works.

## 2. Materials and methods

### 2.1. Synthesis of HAp powders

Materials included: analytical grade (AR) calcium nitrate [$Ca(NO_3)_2$], and ammonium phosphate [$(NH_4)H_2PO_4$]. Solution of aqueous ammonia was used to maintain pH and deionized water was used in all synthesis steps.

Calcium nitrate [$Ca(NO_3)_2$], and ammonium phosphate [$(NH_4)H_2PO_4$] were weighed according to the required stoichiometry (Ca/P molar ratio: 1.67). Calcium nitrate [$Ca(NO_3)_2$], and ammonium phosphate [$(NH_4)H_2PO_4$] were dissolved in deionized water separately. The calcium nitrate solution was vigorously stirred, and the ammonium phosphate solution was added dropwise. The pH of each solution was adjusted to 11 using an solution of aqueous ammonia. Stirring for 1 h produced a milky, somewhat gelatinous precipitate. The precipitate was filtered using a funnel and filter paper. After filtration, it was dried at 80 °C, 120 °C, and 160 °C for 24 h for HAp-1, HAp-2, and HAp-3, respectively. For samples HAp-2 and HAp-3 glutamic acid was added additionally added in the following concentrations 3 M and 5 M respectively. The powders were crushed with a mortar and pestle. In Table 1, the synthesis parameters are shown. The dry powders were crushed using a mortar and taken for characterization. Fig. 1 and Fig. 2 explain the detailed procedure and experimental conditions.

## 3. Characterization of powders

### 3.1. Phase composition: X-ray diffraction (XRD)

X-ray powder diffraction was used to identify the crystalline phases contained in all samples. Wide-angle X-ray experiments were carried out in a Rigaku Dmax 2100 diffractometer using the $CuK_\alpha$ radiation ($\lambda$ = 1.5406 Å), an accelerating voltage of 30 kV and a current of 20 mA. Diffractograms were recorded from 5 through 80 ° on a 2θ scale with a rate of 10 ° per minute. Spectrum analysis software, MDI Jade V 5.0.37, was used [17].

### 3.2. Morphology and microstructure: Scanning electron microscopy (SEM)

Morphological, topological, and microstructural analyses of all samples were carried out in a JEOL JXA-8530F Scanning Electron Microscope. The analysis was performed using 20 kV electron acceleration voltage, and secondary electrons formed the images. All the samples were placed on a stainless-steel plate with a separation of 5 mm and pasted with silver paint and covered with a gold thin film by sputtering to avoid the electrostatic charge accumulation [17].

### 3.3. Elemental composition: X-ray fluorescence (XRF)

To obtain the elemental composition and the Ca/P ratio of hydroxyapatite powders, a Bruker S2 Picofox was used. The operating conditions were as follows: voltage of 50 kV and $MoK_\alpha$ radiation. The samples were dispersed in 1 mL of isopropyl alcohol and iron was added as an internal standard [18].

## 4. Results and discussion

### 4.1. X-ray diffraction (XRD)

X-ray diffraction is used to analyze the crystalline structure of matter. Characteristics such as diffraction angle and diffracted beam intensity help to identify the structure and determine

**Table 1** Experimental conditions in the synthesis process of HAp powders.

| Sample | Precursors | Drying temperature °C |
|---|---|---|
| HAp-1 | $Ca(NO_3)_2 + (NH_4)H_2PO_4$ | 80 |
| HAp-2 | $Ca(NO_3)_2 + (NH_4)H_2PO_4$ + Glutamic Acid | 120 |
| HAp-3 | $Ca(NO_3)_2 + (NH_4)H_2PO_4$ + Glutamic Acid | 160 |

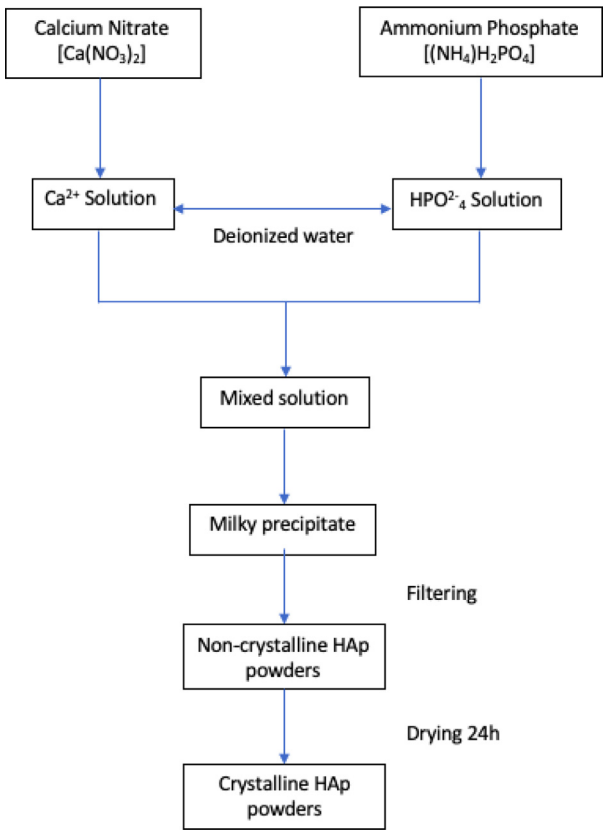


**Fig. 1** Modified precipitation synthesis process for HAp powders (Adapted from Monmaturapoj [16]).

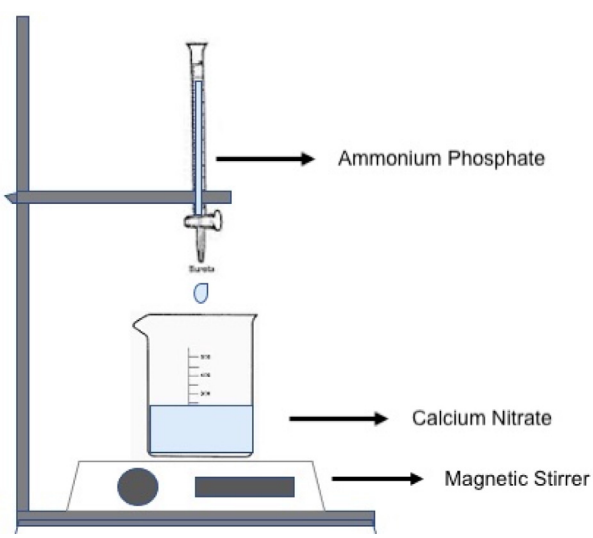


**Fig. 2** Schematic representation of the synthesis of HAp powders using precipitation method.

structural parameters of the material such as preferred orientation, crystalline planes, and crystallite size [19].

An XRD analysis for all HAp powders obtained was carried out. In Fig. 3 the diffraction patterns for HAp-1, HAp-2 and HAp-3 are shown. The results demonstrate that all HAp powders were pure crystalline products. It was observed that XRD identification phase of HAp synthesized powders exactly match with the XRD HAp pattern compared to the diffraction pattern with Iinternational Centre for Diffraction Data (ICDD) Powder Diffraction File (PDF) # 01–084-1998. The most relevant reflections observed in the diffractograms correspond to (100), (002), (300) and (222) planes present in all samples and relative intensities vary from sample to sample.

The intensity ratio of the two most intense Bragg reflections was calculated from the mentioned PDF HAp-1, HAp-2, and HAp-3. For HAp-1 the (002)/(300) intensity ratio according to the PDF is 0.58, while for HAp-1 powders obtained were 0.51. Similarly, for HAp-2, the (100)/(002) intensity ratio in PDF is 0.49 while in the sample it is 0.48, finally, for HAp-3, the (112)/(300) intensity ratio is 0.85 and 0.76 for the PDF and the sample, respectively. A texture index was established using the whole patterns resulting in an intensity ratio of 0.64.

The crystallite size (D) of the HAp powders was calculated, using the whole patterns, from the X-ray diffraction data using the Scherrer equation [20]. In this Eq. (1), K is the Scherrer's constant, which is taken as 0.94, λ presents the wavelength of CuKα, B is for the full width at half maximum (FWHM) and θ stands for the diffraction angle. The results are shown in Table 2.

$$D = \frac{K \cdot \lambda}{\beta \cdot cos\theta} \tag{1}$$

The HAp powders heat-treated at high temperature and adding glutamic acid as a precursor (HAp-2 and HAp-3) exhibited sharp and narrow peaks compared to HAp-1 powders treated at 80 °C without glutamic acid. According to other works, it is believed that the grain growth and the improvement of crystallinity due to the increase in temperature in the heat treatment is the most influential factor in this situation [12]. However, the presence of glutamic acid in the synthesis process plays an essential role in the preferential growth of crystals. The thin and narrow peaks are signs of well-developed crystals. As shown in Table 2 and according to Scherrer's equation the crystallite size increases as the peak width decreases.

### *4.2. Scanning electron microscopy (SEM).*

The morphology, surface characteristics as well as crystallographic information of synthesized powders HAp were analyzed from the SEM results.

The SEM micrographs of the synthesized HAp powders were compared with the same magnification X500, X1000, and X2000. In Fig. 4 shows HAp-1 micrograph, in this case plaque-shaped hydroxyapatite agglomerates are observed. The morphology is not well defined, and the agglomerates have a rectangular shape with dimensions in the order of 200 μm long, 150 μm wide and 30 μm thick. Smaller agglomerates formed by sheets are observed on the surface.

In Fig. 5, plate-shaped crystals were observed, the uniform grain size was closely related to the improvement in the crystallinity of the HAp-2 powders when they were calcined at a higher temperature (120 °C). Compared to the HAp-1 sample, the morphology of the plates is well defined with smaller dimensions.

Fig. 6 shows the micrographs for HAp-3; in the images, plate-shaped crystals with dimensions in the order of 10 nm

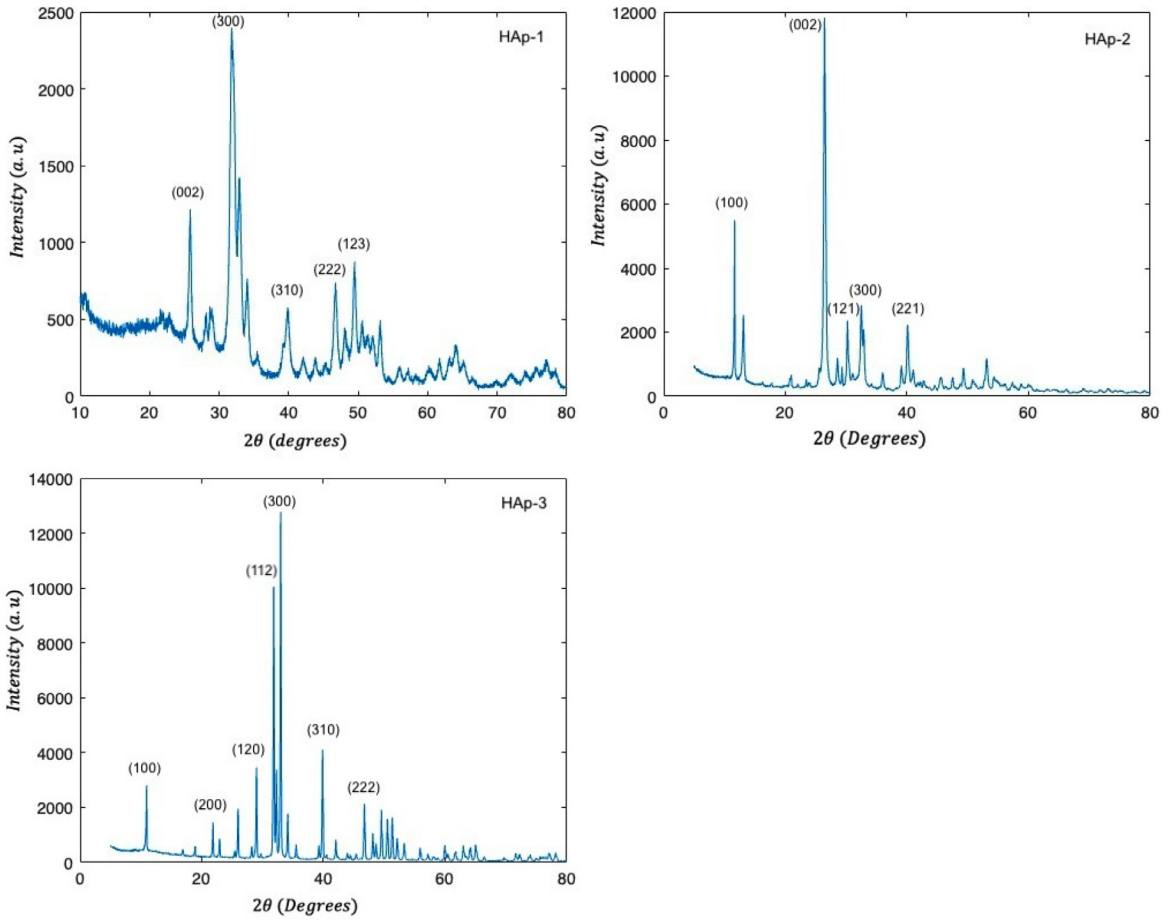


**Fig. 3** XRD spectra of the HAp powders synthesized using precipitation method.

**Table 2** Mean crystallite size of HAp powders.

| Sample | Wavelength (Å) | Peak Width (°) | Crystallite Size (nm) |
|---|---|---|---|
| HAp-1 | 1.5406 | 0.34 | 20.13 |
| HAp-2 | 1.5406 | 0.19 | 52.64 |
| HAp-3 | 1.5406 | 0.17 | 60.31 |

are observed. Additionally agglomerates of hydroxyapatite in the form of a rod can be observed, the presence of glutamic acid added to the increase in the calcination temperature suggest the change in the morphology of some crystals present in the sample. According to other research, the quality, purity, morphology, and particle size depending on the synthesis parameters involved in the precipitation reaction, such as temperature, pH, Ca / P molar ratio, and ionic contaminants [21]. The well-defined morphology of the microplates appears with a gradual increase in temperature from 80, 120 and 160 °C which indicates that the high temperatures were useful to improve the crystallinity of the HAp powders obtained by precipitation. Since the wet precipitation process is based on the nucleation-growth mechanism, the efficiency of growth and development of the nuclei influences the crystallinity of the material obtained [22]. Numerous reports have shown that synthesis at low temperatures leads to HAp crystals with irregular morphology, and the fact that the particles show a greater

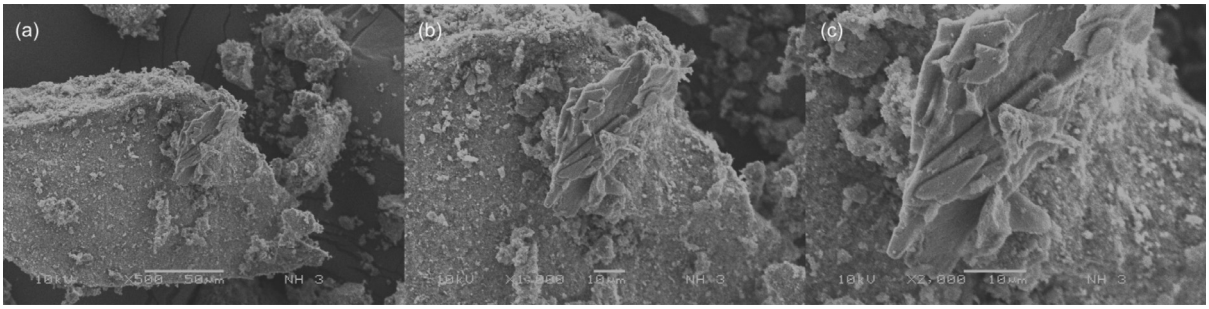


**Fig. 4** SEM images of the HAp-1 (a) X500, (b) X1000 and (c) X2000.

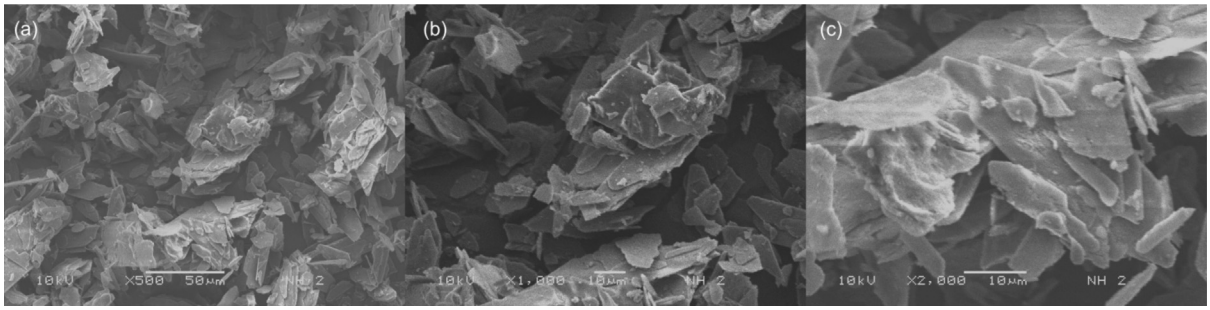


**Fig. 5** SEM images of the HAp-2 (a) X500, (b) X1000 and (c) X2000.

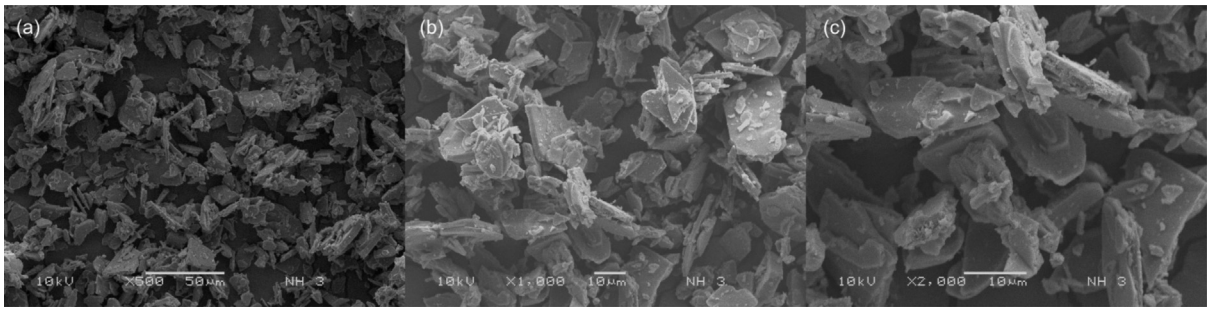


**Fig. 6** SEM images of the HAp-3 (a) X500, (b) X1000 and (c) X2000.

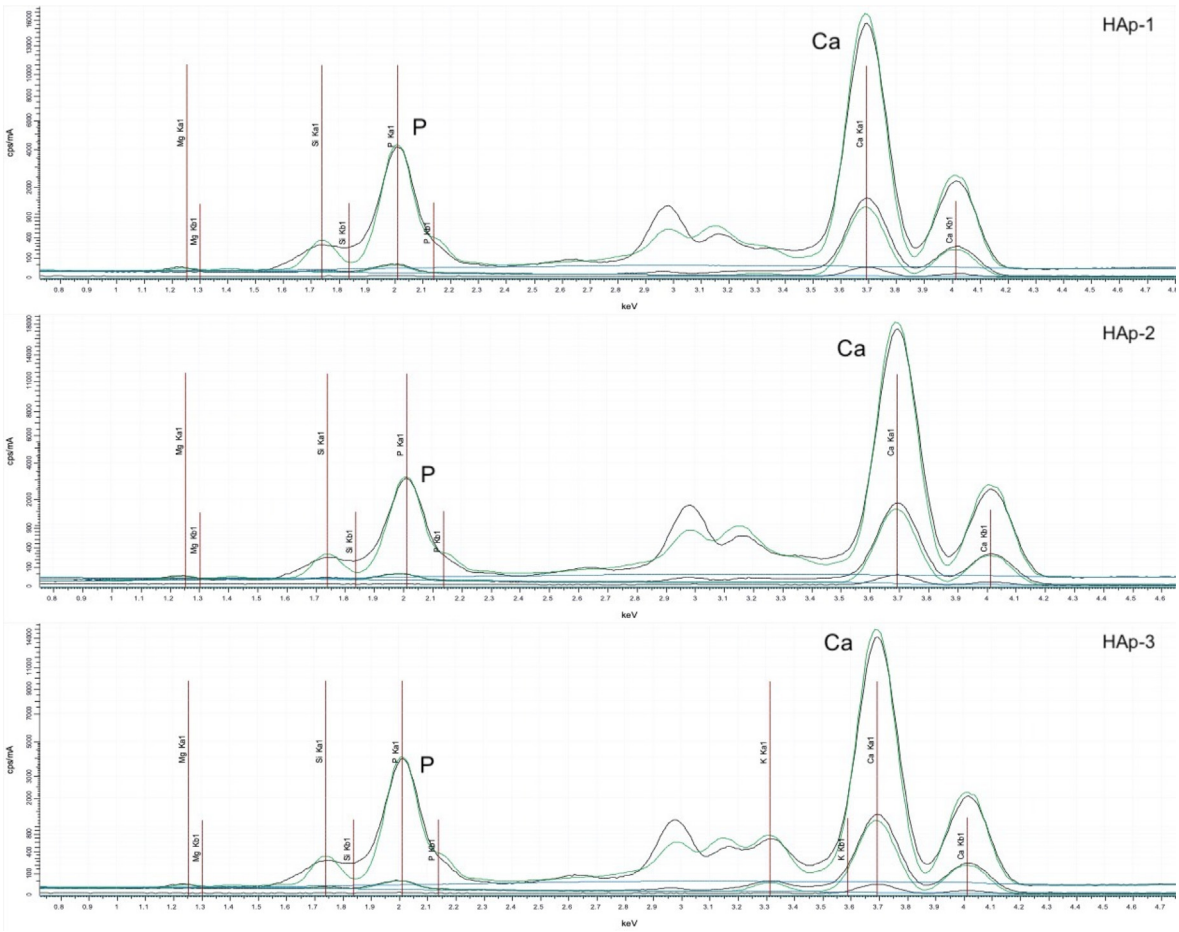


**Fig. 7** XRF spectra of HAp-1, HAp-2 and HAp-3.

inclination to agglomerate [23]. Therefore, it is suggested that the transition from irregular to regular morphologies and the increase in reaction temperature coincides with the increase in crystallinity in the HAp-2 and HAp-3 samples [15].

Previously, Legeros et al. [24] reported that the hydroxyapatite produced through the aqueous system by precipitation is usually deficient in calcium and enriched in $HPO_4$ according to the axis "a" dimensions crystalline structure, which agrees with the results obtained by XRD and SEM. In agreement with the XRD results, the diffraction plane in the direction of the "a" axis suggests that the hydroxyapatite powders are deficient in calcium. According to SEM observations, the plaque morphology is due to this calcium deficiency and the presence of glutamic acid.

### 4.3. X-ray fluorescence (XRF)

XRF shows the elemental composition of the compounds present in the material. Calcium and phosphorus are the primary precursors in the synthesis of hydroxyapatite and their presence can be seen present in Fig. 7. Other impurities are observed in the XRF spectrum such as Magnesium and silicon dioxide but in very low percentages about calcium, and phosphorus.

Molar ratio values were determined according to the data obtained by XRF. The stoichiometric (Ca/P) molar ratios present in the samples were 1.78, 1.77 and 1.75 for HAp-1, HAp-2 and Hap-3 respectively. These results agree with the stoichiometric ratio of hydroxyapatite (Ca/P = 1.67). It is understood that although the heat treatment carried out at 80 °C is successful to obtain HAp powders, 160 °C as the heat treatment temperature is more effective to ensure the molar ratio (Ca/ P) in all the powders [12]. This situation suggests a better crystallization at 160 °C, this is due to the mechanism of growth and development of the grain that depends on the temperature of the heat treatment. On the other hand, the addition of glutamic acid showed a tendency to increase the particle size and its preferential growth direction.

## 5. Conclusions

The wet chemical precipitation technique produces HAp powders on the order of microns. The process parameters in the wet chemical precipitation method can be varied to improve the properties of the obtained product. XRD and SEM studies confirmed the formation of hydroxyapatite microplates and microfibers. HAp powders heat-treated at 80 °C have a smaller crystallite size than to HAp powders heat-treated at 120 °C. Nevertheless, HAp-1, HAp-2 and HAp-3 powders can be indicated as a candidate material used in three-dimensional (3d) printing systems through a simple grinding process after heat treatment. XRF studies confirmed a similar stoichiometry to natural hydroxyapatite, therefore suggesting possible biomedical applications in the regeneration of bone tissue. Previously, ceramic powders derived from aqueous methods have been reported to have a porous morphology. Although porosity is an unfavorable property in terms of mechanical properties. It also helps the material to be biochemically compatible with tissues. Our previous investigations [18] based on investigating different calcium phosphates showed good results in obtaining ceramic powders with possible biomedical applications. Glutamic acid plays an important role in the preferential growth of structures and is a parameter to consider in the synthesis process.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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